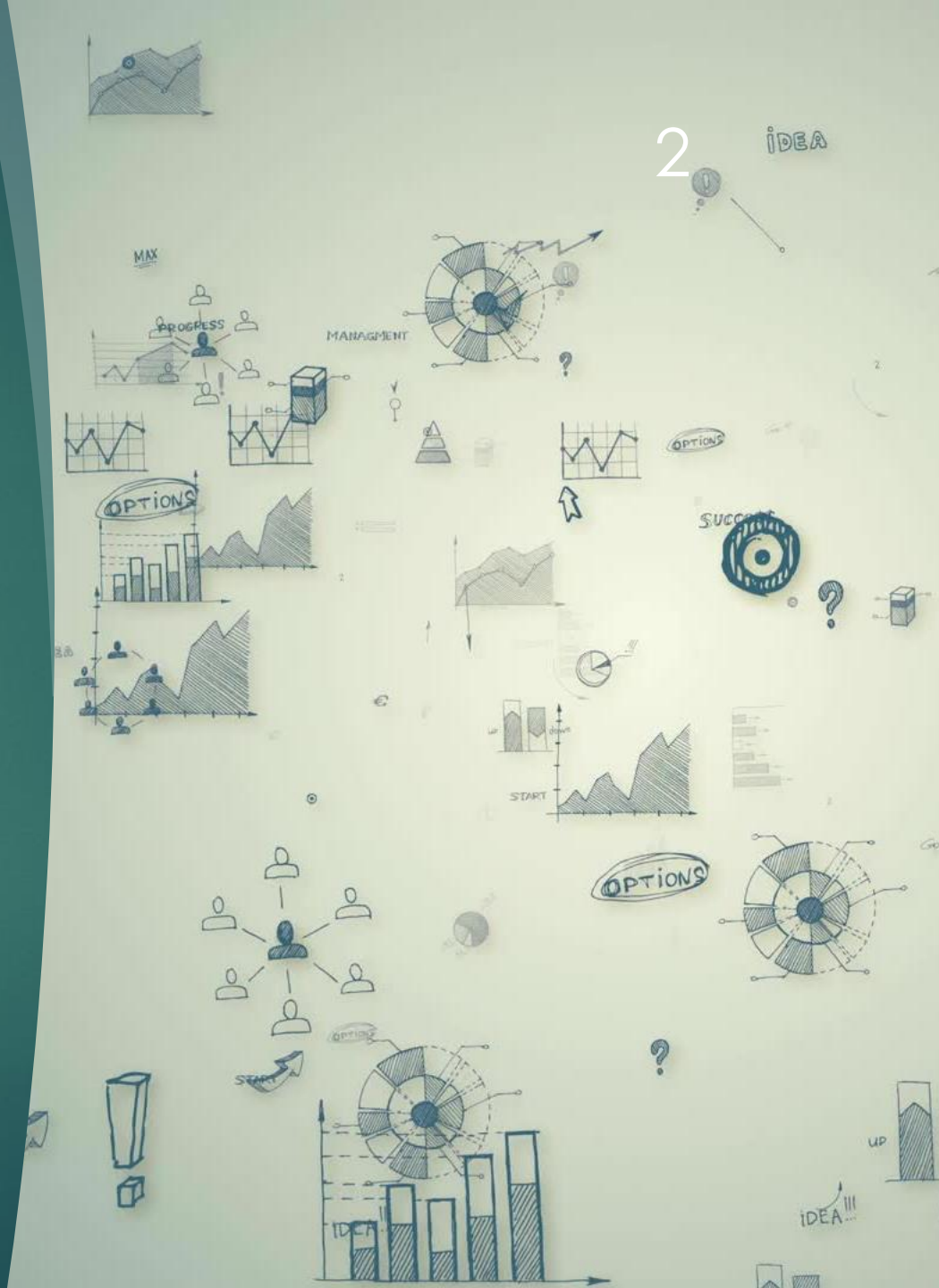
Introduction

CHAPTER 1

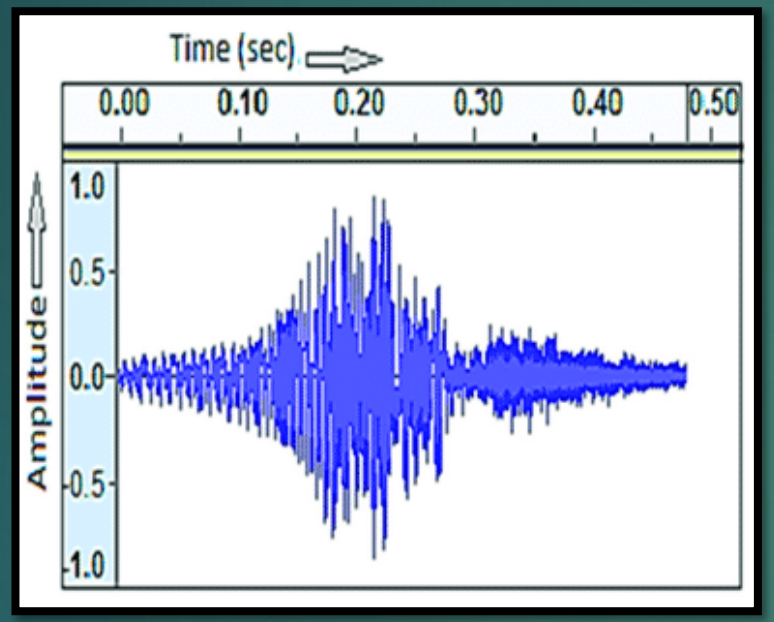
AN ACCESSIBLE APPROACH TO PATTERN RECOGNITION

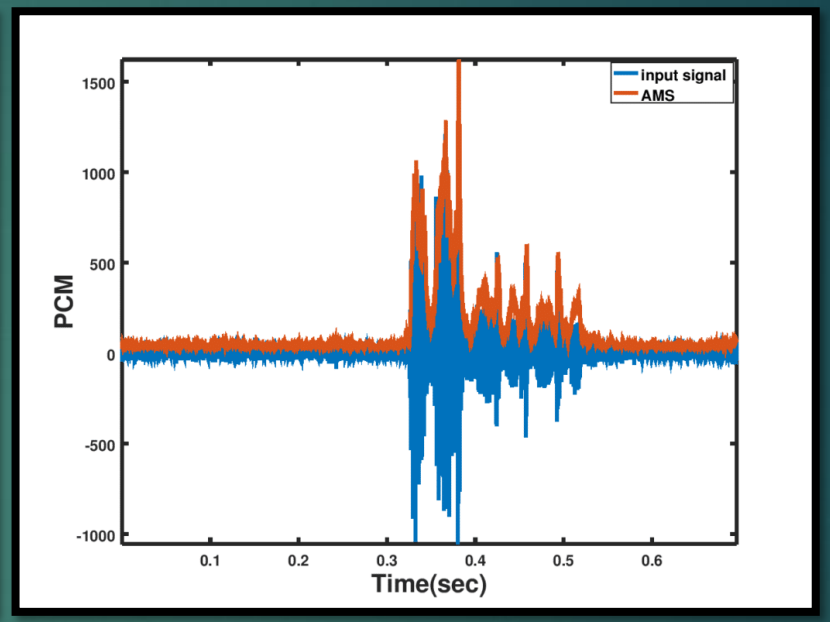
🔍 Chapter 1 Roadmap

- Autonomous driving example
- What is pattern recognition?
- Relationship with machine learning
- The big picture of a PR system.



Pattern Recognition





Introduction

- Pattern Recognition is:
  - Teaching computers to learn from examples — **from autonomous driving 🚗 to medical diagnosis 🏥**
- 🤖 Real-world hook :
  - How does a self-driving car recognize a pedestrian?
  - How does your phone unlock with your face?
  - This is pattern recognition in action!

🚗 An Example: Autonomous Driving

- 🚦 **The Challenge**
  - A self-driving car must perceive its environment in real-time. It needs to recognize:
    - 🚶 Pedestrians crossing the street
    - 🚗 Other vehicles and their trajectories
    - 🛑 Traffic signs and signals
    - 🐕 Unexpected obstacles (animals, debris)
- 🤖 **The Pattern Recognition Solution**
  - The car's computer processes camera images, extracts **features** (edges, textures, shapes), and uses a **classifier** to label objects in milliseconds.
  - ⚡ **Speed matters:** A pedestrian stepping into the road gives the car less than a second to react. Pattern recognition must be both **accurate** and **fast**.

🧠 Pattern Recognition and Machine Learning

- 🎯 **Definition**
  - **Pattern recognition** is the scientific discipline concerned with the **automatic discovery of regularities in data** and the use of these regularities to make decisions (classify new observations).
- 📚 **Two Learning Paradigms**
  - ✓ **Supervised learning:** Learning from labeled examples (e.g., teaching a computer to recognize cats vs. dogs using labeled photos).
  - ✓ **Unsupervised learning:** Finding hidden structures without labels (e.g., grouping customers by purchasing behavior).
- 🔗 **Relationship with ML**
  - Pattern recognition is the **application side** of machine learning.
  - While ML focuses on algorithms that learn from data, **pattern recognition focuses on the end-to-end system** that solves real-world tasks.

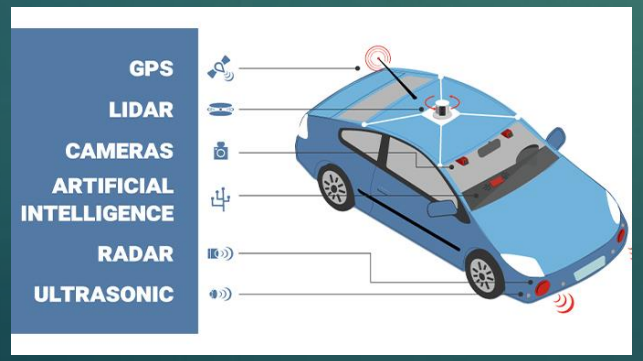
Why Study Pattern Recognition in 2026?

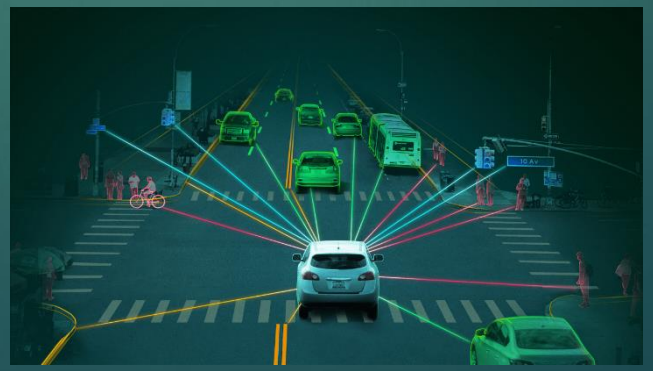
- 📱 **Everyday Applications**
- Face unlock on your phone 📱
- Spam filters in Gmail 📧
- Speech assistants (Siri/Alexa) 🎤
- Recommendation systems (Netflix/Amazon) 📺

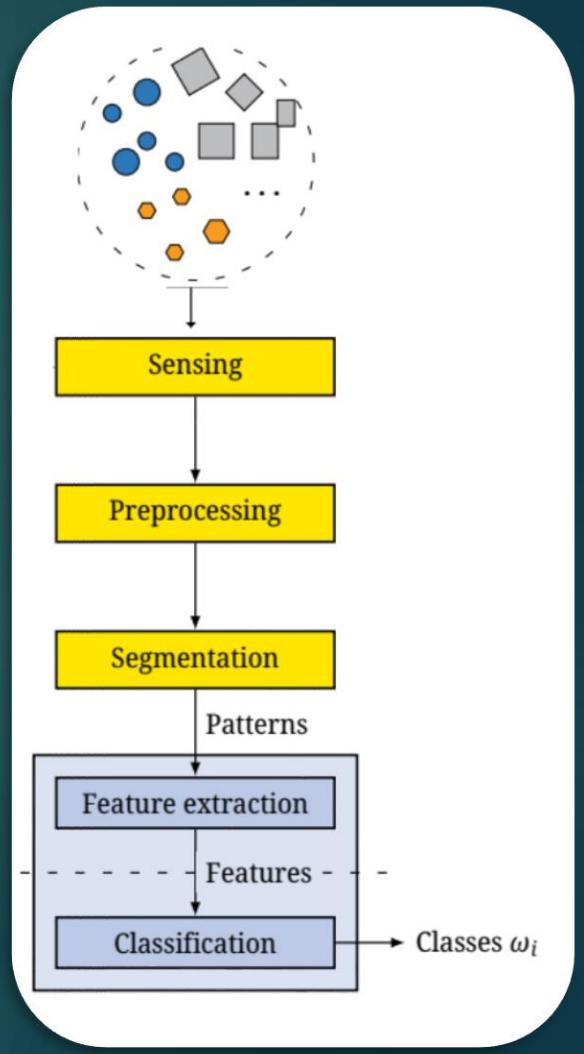
- 🏥 **Life-Changing Applications**
- Medical image diagnosis (cancer detection)
- Drug discovery 💊
- Climate change modeling 🌍
- Assistive technology for disabilities ♿

The Big Picture: A Pattern Recognition System

- 🚗 **Autonomous driving example in this framework:**
  - **Sensors:** Cameras, LiDAR, radar → raw pixels/point clouds
  - **Features:** Edge orientations, depth maps, object proposals
  - **Classifier:** Deep neural network trained on millions of labeled road scenes
  - **Output:** "Pedestrian at (x,y)" → brakes applied!







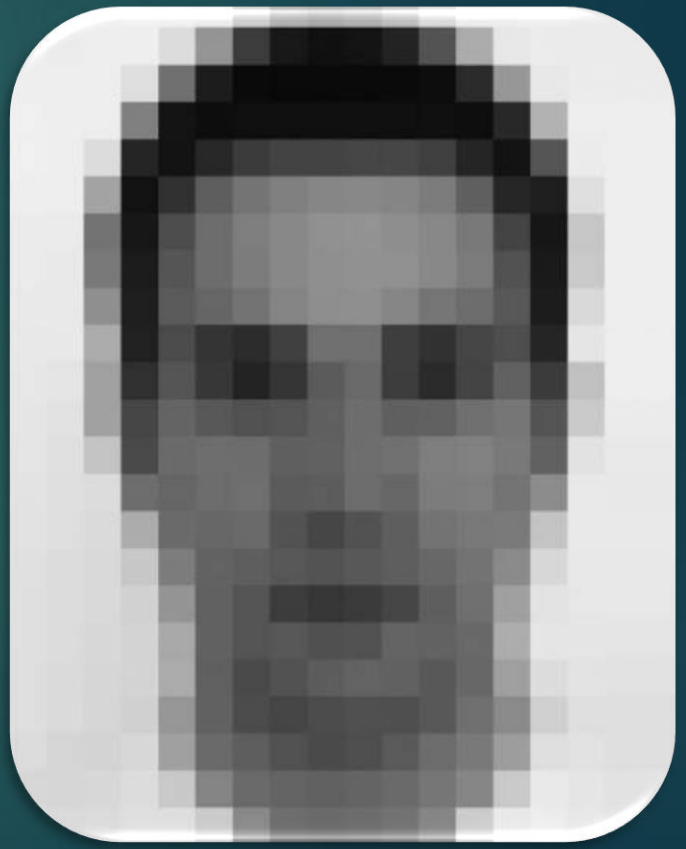
The Curse of Dimensionality

- **Bellman's Curse**
  - As the number of features (dimensions) grows, the volume of space grows *exponentially*.
  - Data becomes increasingly sparse, distances become meaningless, and you need exponentially more training samples.
  - This is why dimensionality reduction techniques (PCA, LDA) are so important.
- 📊 **Real Example: High-Dimensional Genomics**
  - Gene expression datasets may have **20,000 features (genes**) but only **200 patient** samples. Direct classification fails catastrophically.
  - Scientists must first apply **dimensionality reduction** — selecting the few hundred genes most relevant to the disease — before any classifier can work effectively.
  - This is the curse of dimensionality in practice.

FEATURES AND FEATURE VECTORS

- **Where systems fail?**
  - Most real-world PR failures happen in the **sensing** or **feature extraction** stages — not in the classifier itself.
  - Poor sensor data and uninformative features **cannot be rescued by a sophisticated algorithm.**
- A **feature** is a measurable property of the phenomenon being observed.
- Selecting the right features is often the single most impactful decision in building a PR system.

"Give a smart classifier poor features and it will fail. Give a simple classifier great features and it will succeed."



Classifiers Overview

📐 Geometrical / Distance-Based

- Classify based on distance to class prototypes or boundaries in feature space. Nearest Neighbor is the classic example.

🎲 Statistical / Probabilistic

- Use probability theory and Bayes' rule to find the class with highest posterior probability given the observation.

🧠 Neural / Learning-Based

- Inspired by biological neurons. Learn complex boundaries through training. Basis of modern deep learning.

The Minimum Distance Classifier

- The simplest geometrical classifier. Assign an unknown pattern to the class whose **prototype (mean vector)** it is closest to.
- 🍎🍊 **Real Example: Fruit Classification**
  - We measure two features per fruit: **weight (grams)** and **redness (0–255)**. Training data tells us the Apple mean is (182g, 220 red) and Orange mean is (130g, 170 red). A new fruit at (175g, 210 red) is closer to the Apple prototype → classified as 🍎 Apple. Simple, fast, surprisingly effective for well-separated classes!

🎲 Bayesian Classification — Think in Probabilities

- Bayes' theorem lets us reason about class membership under uncertainty.
- It combines **prior knowledge** (how common is each class?) with **likelihood** (how probable is this observation given the class?).

$$P(\text{class} \mid x) = P(x \mid \text{class}) \times P(\text{class}) / P(x)$$

$$\text{Posterior} = \text{Likelihood} \times \text{Prior} / \text{Evidence}$$

- 🩺 **Real Example: COVID Test Interpretation**
  - **Prior:** 1% of people in a region are infected.
  - **Likelihood:** Test is 95% accurate if infected, 3% evidence.
  - **A positive test result:** Using Bayes → only ~32% chance you're actually infected!
  - This counterintuitive result (because prior infection rate is low) is why doctors consider prevalence when interpreting tests — pure Bayesian reasoning.

🧠 Neural Network Intuition

- Neural networks learn to classify by adjusting **weights** through training examples. Each neuron computes a weighted sum of inputs and applies a nonlinear activation — stacked layers allow learning of increasingly abstract features.
- **Why Neural Networks?**
- For raw images, audio, or text, manually engineering good features is extremely hard. Neural networks learn *both* the features *and* the classifier jointly, end-to-end — this is their superpower.

Real-World Applications

🚗 Autonomous Vehicles

- Tesla's **Autopilot** classifies road markings, traffic signs, pedestrians, and vehicles in real-time from camera and lidar data.

💳 Fraud Detection

- Visa processes 65,000 transactions/second, using PR to flag unusual spending patterns within milliseconds — saving billions annually.

🗣 Speech Recognition

- Siri, Alexa, and Google Assistant convert acoustic waveforms to text using deep neural networks trained on thousands of hours of speech.

🔬 Medical Imaging

- AI systems match or exceed radiologist accuracy in detecting diabetic retinopathy from eye scans

🌿 Agriculture

- Drone cameras scan fields and classify plant diseases from leaf images

📱 Face Unlock

- Your phone's Face ID maps 30,000 infrared dots onto your face geometry and classifies you among billions of possible identities